



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Class Activity – 24/07/2026

Name: Jana Lahari

Reg No: 192424224

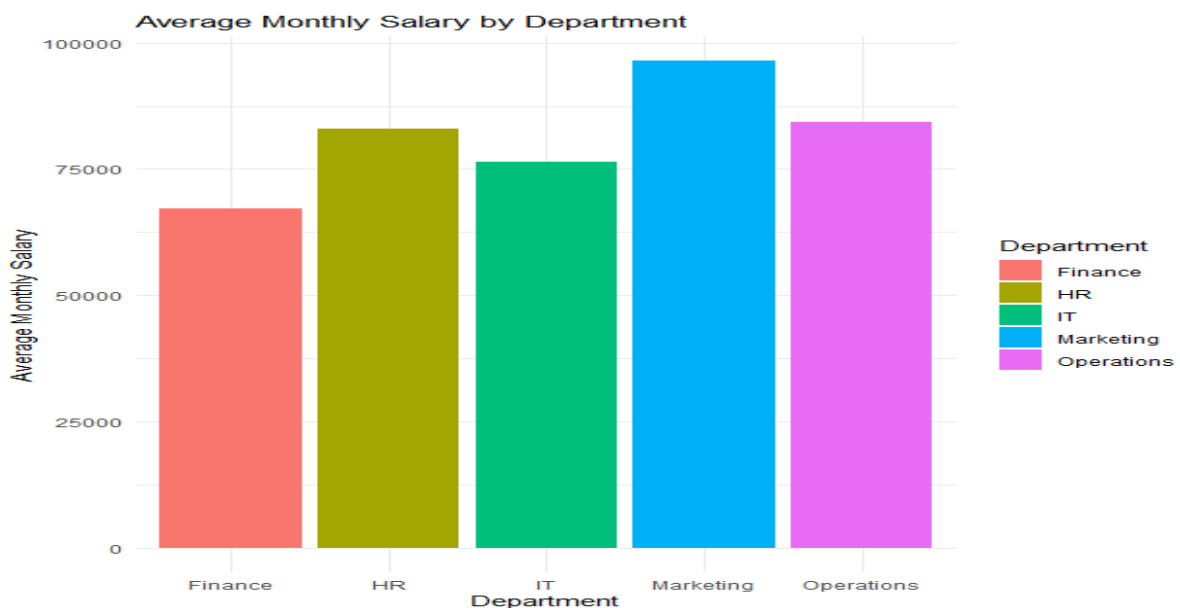
Employee Salary Visualizations

1.Bar chart

R Code:

```
bar_data <- aggregate(Monthly_Salary ~ Department, employee, mean)
ggplot(bar_data,
  aes(x = Department,
    y = Monthly_Salary,
    fill = Department)) +
  geom_bar(stat = "identity") +
  labs(title = "Average Monthly Salary by Department",
    x = "Department",
    y = "Average Monthly Salary") +
  theme_minimal()
```

Output:

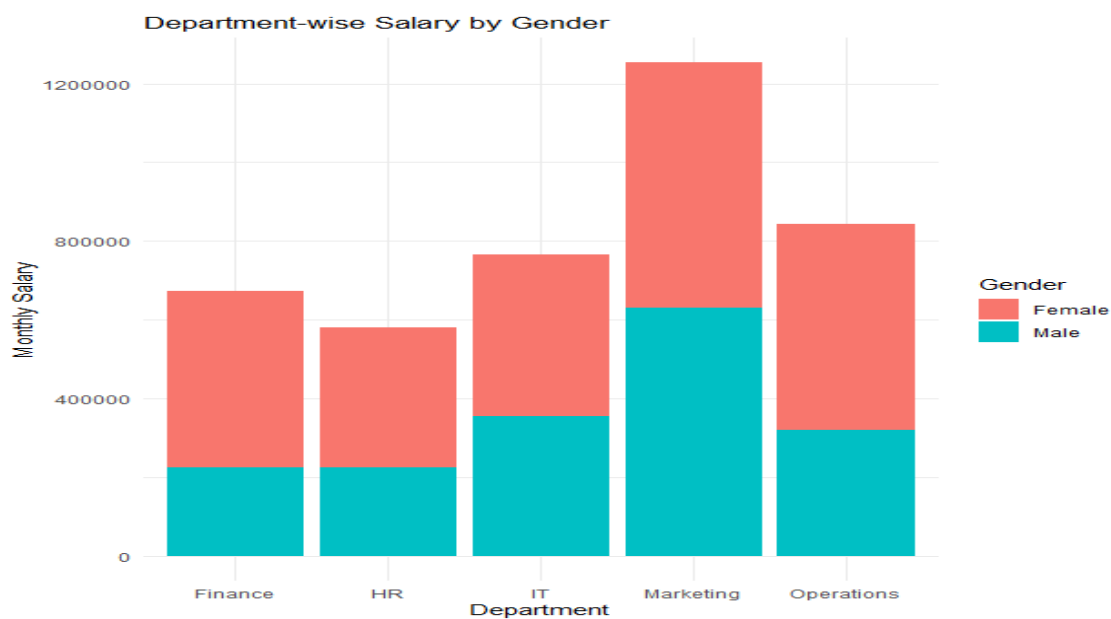


2.Stacked bar chart

R Code:

```
ggplot(employee,  
  aes(x = Department,  
      y = Monthly_Salary,  
      fill = Gender)) +  
geom_bar(stat = "identity") +  
labs(title = "Department-wise Salary by Gender",  
  x = "Department",  
  y = "Monthly Salary") +  
theme_minimal()
```

Output:

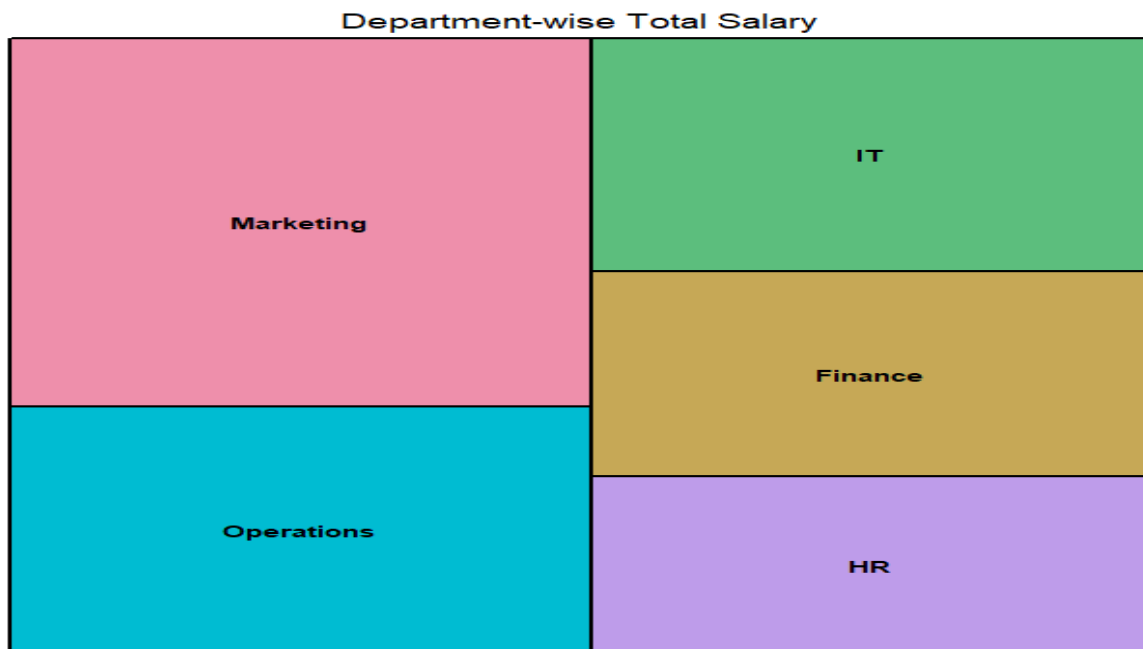


3.Tree Map

R Code:

```
tree_data <- aggregate(Monthly_Salary ~ Department, employee, sum)  
treemap(tree_data,  
  index = "Department",  
  vSize = "Monthly_Salary",  
  title = "Department-wise Total Salary")
```

Output:

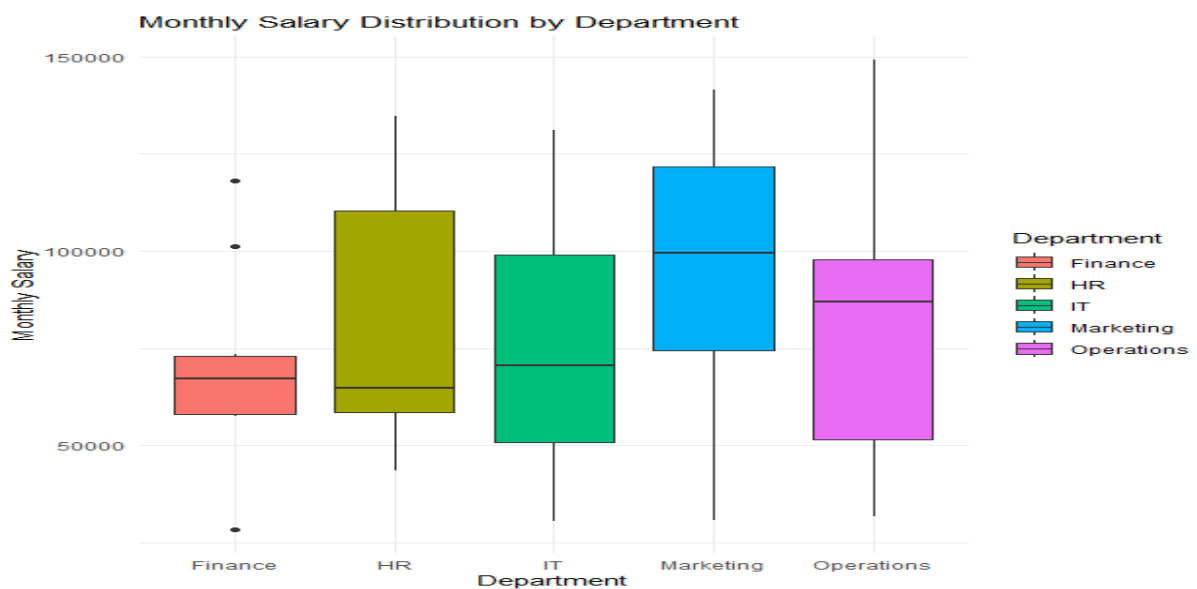


4.Box plot

R Code:

```
ggplot(employee,  
  aes(x = Department,  
      y = Monthly_Salary,  
      fill = Department)) +  
geom_boxplot() +  
labs(title = "Monthly Salary Distribution by Department",  
  x = "Department",  
  y = "Monthly Salary") +  
theme_minimal()
```

Output:

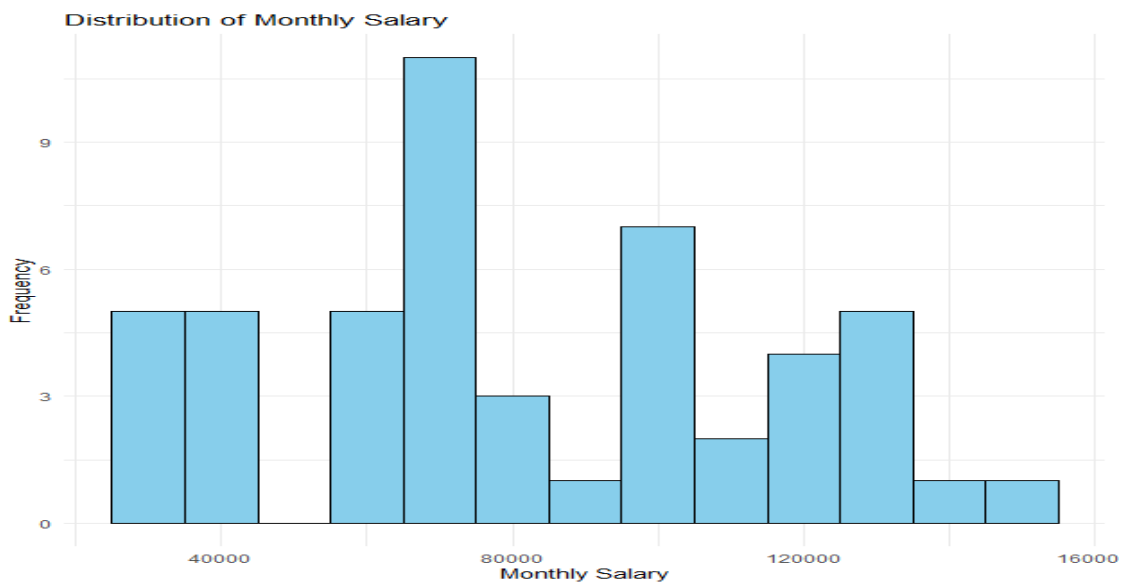


5.Histogram

R Code:

```
ggplot(employee,  
  aes(x = Monthly_Salary)) +  
  geom_histogram(binwidth = 10000,  
    fill = "skyblue",  
    color = "black") +  
  labs(title = "Distribution of Monthly Salary",  
    x = "Monthly Salary",  
    y = "Frequency") +  
  theme_minimal()
```

Output:

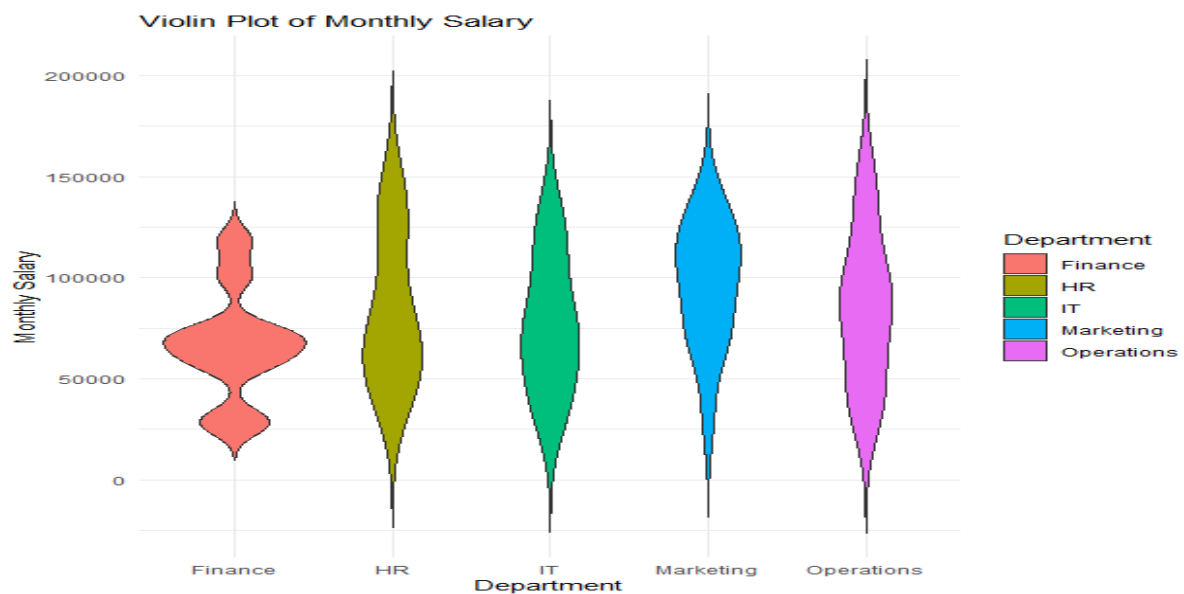


6.Violin plot

R Code:

```
ggplot(employee,  
  aes(x = Department,  
    y = Monthly_Salary,  
    fill = Department)) +  
  geom_violin(trim = FALSE) +  
  labs(title = "Violin Plot of Monthly Salary",  
    x = "Department",  
    y = "Monthly Salary") +  
  theme_minimal()
```

Output:

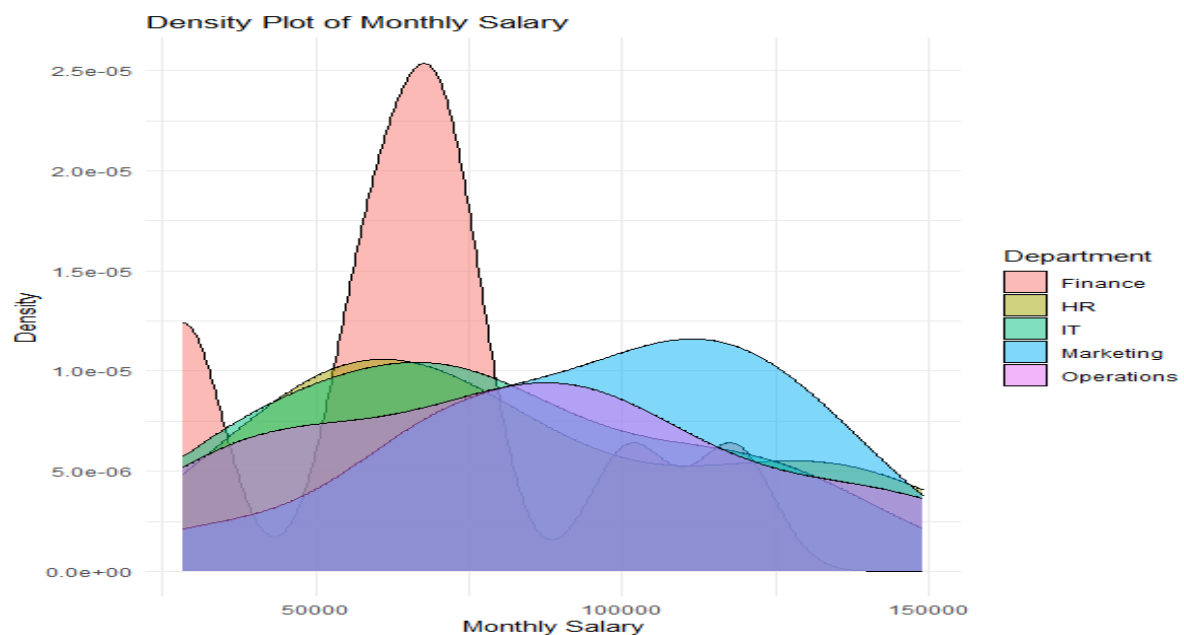


7. Density plot

R Code:

```
ggplot(employee,  
  aes(x = Monthly_Salary,  
    fill = Department)) +  
  geom_density(alpha = 0.5) +  
  labs(title = "Density Plot of Monthly Salary",  
    x = "Monthly Salary",  
    y = "Density") +  
  theme_minimal()
```

Output:

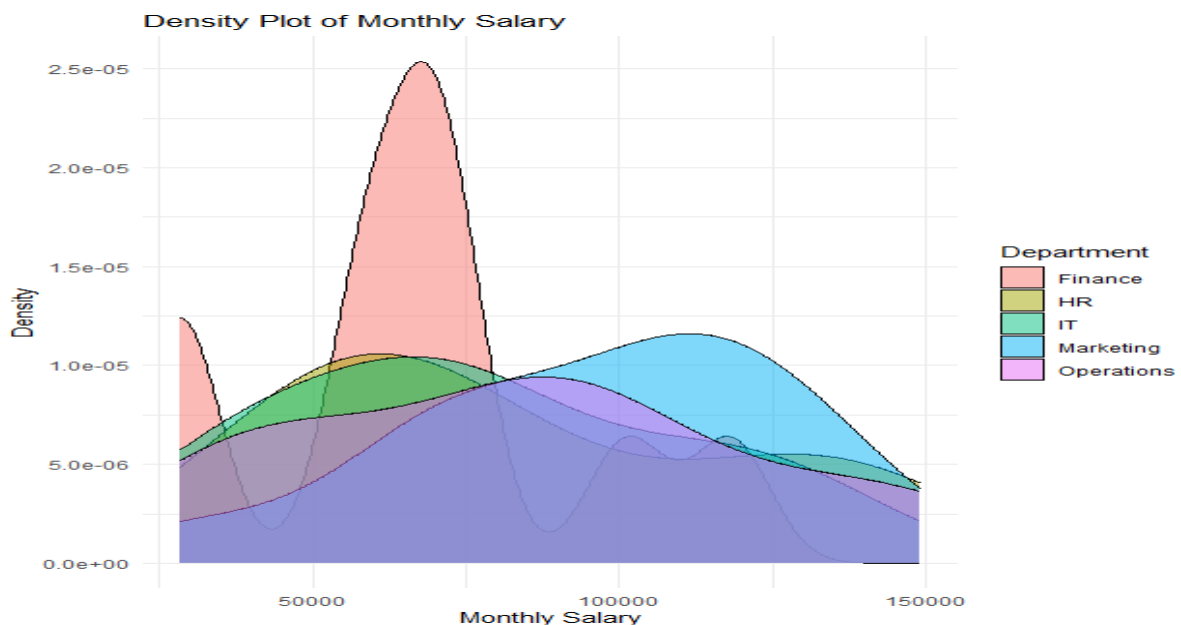


8.Scatter plot

R Code:

```
ggplot(employee,  
  aes(x = Experience_Years,  
      y = Monthly_Salary,  
      color = Department)) +  
geom_point(size = 3) +  
labs(title = "Experience vs Monthly Salary",  
  x = "Experience (Years)",  
  y = "Monthly Salary") +  
theme_minimal()
```

Output:



9.Heat map

R Code:

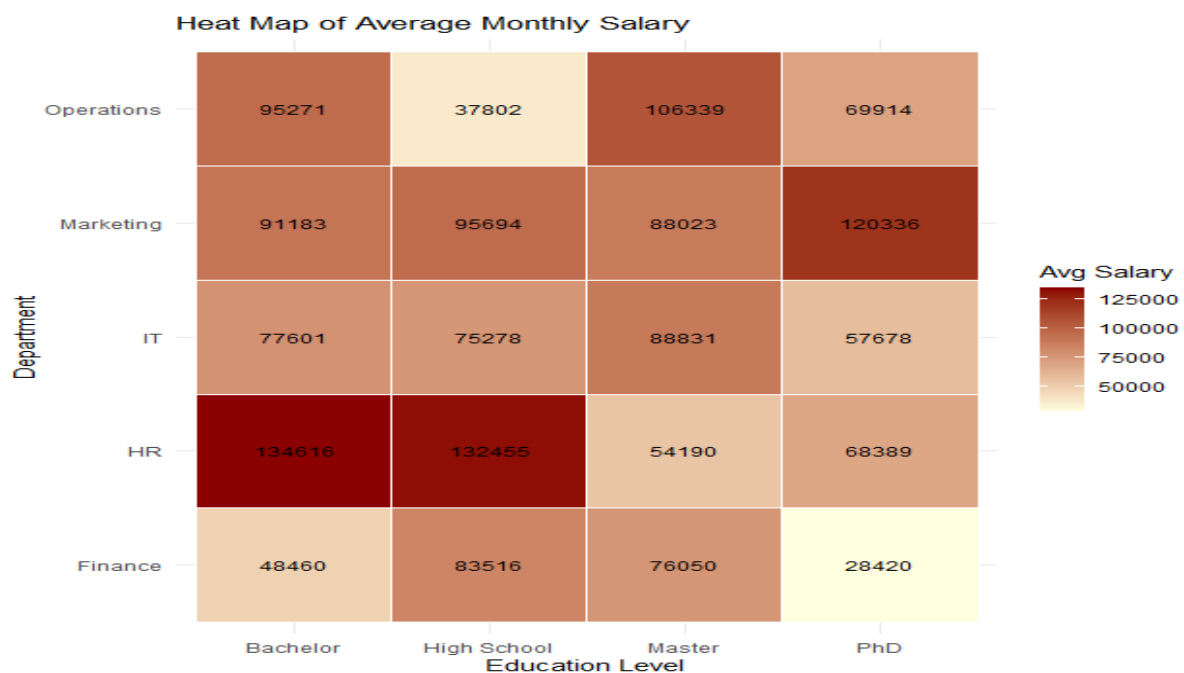
```
heat_data <- employee %>%  
  group_by(Department, Education_Level) %>%  
  summarise(Avg_Salary = mean(Monthly_Salary), .groups = "drop")  
ggplot(heat_data,  
  aes(x = Education_Level,  
      y = Department,  
      fill = Avg_Salary)) +  
geom_tile(color = "white") +  
geom_text(aes(label = round(Avg_Salary, 0)),
```

```

    color = "black",
    size = 3) +
scale_fill_gradient(low = "lightyellow",
    high = "darkred") +
labs(title = "Heat Map of Average Monthly Salary",
    x = "Education Level",
    y = "Department",
    fill = "Avg Salary") +
theme_minimal()

```

Output:



10. Bubble chart

R Code:

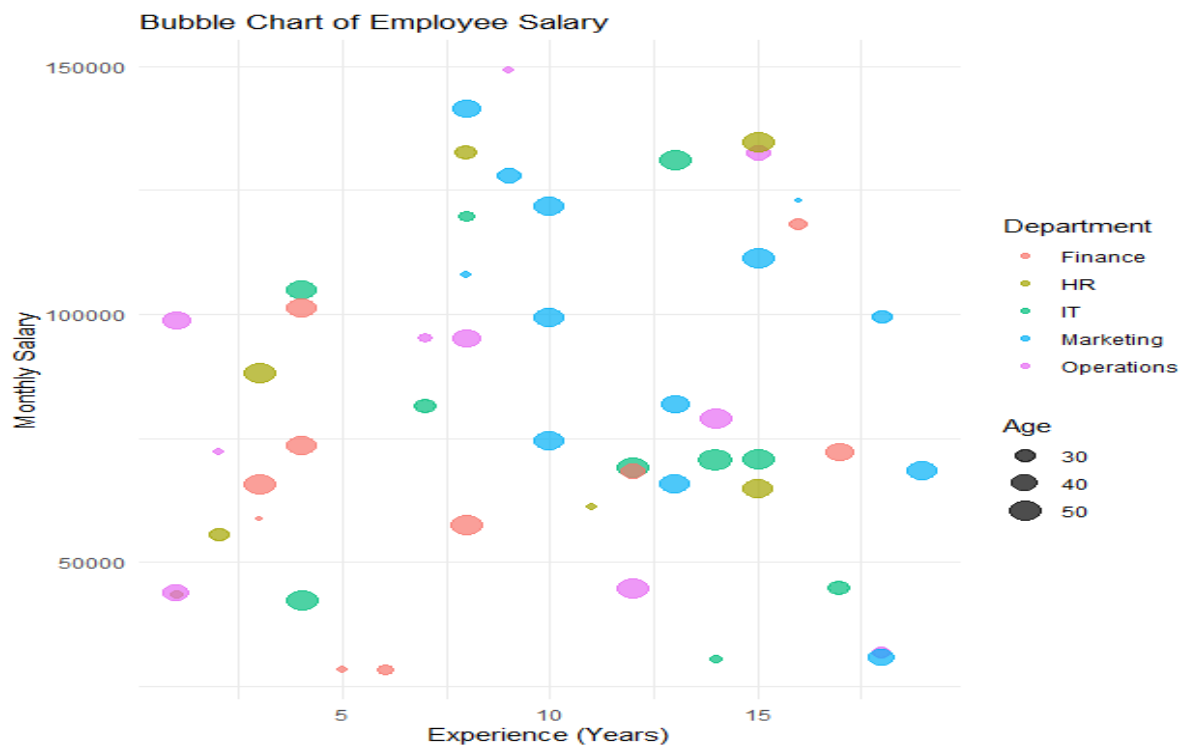
```

library(ggplot2)
ggplot(employee,
    aes(x = Experience_Years,
        y = Monthly_Salary,
        size = Age,
        color = Department)) +
geom_point(alpha = 0.7) +
labs(title = "Bubble Chart of Employee Salary",
    x = "Experience (Years)",
    y = "Monthly Salary",
    size = "Age",

```

```
color = "Department") +
theme_minimal()
```

Output:

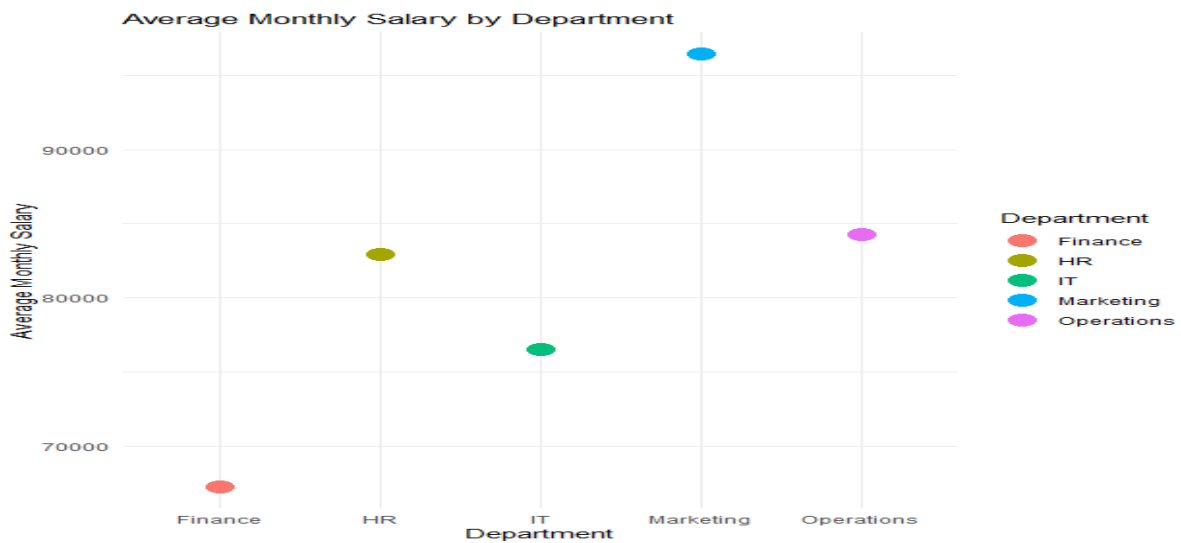


11.Dot chart

R Code:

```
dot_data <- aggregate(Monthly_Salary ~ Department, employee, mean)
library(ggplot2)
ggplot(dot_data,
  aes(x = Department,
    y = Monthly_Salary,
    color = Department)) +
  geom_point(size = 5) +
  labs(title = "Average Monthly Salary by Department",
    x = "Department",
    y = "Average Monthly Salary") +
  theme_minimal()
```


Output:

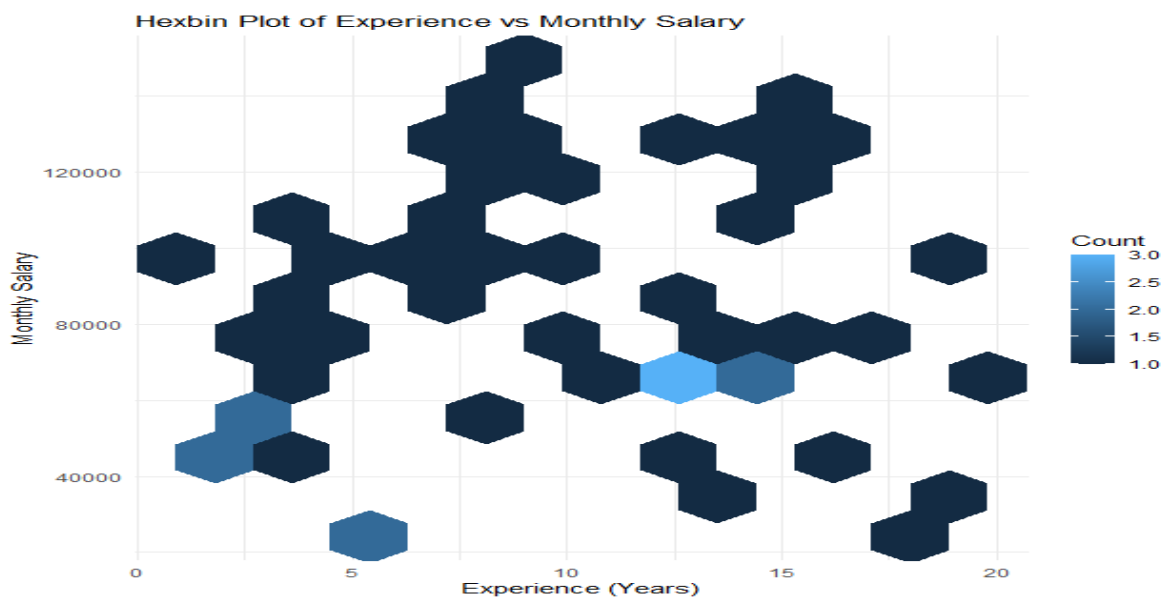


12. Bubble chart

R Code:

```
library(ggplot2)
library(hexbin)
ggplot(employee,
  aes(x = Experience_Years,
      y = Monthly_Salary)) +
  geom_hex(bins = 10) +
  labs(title = "Hexbin Plot of Experience vs Monthly Salary",
       x = "Experience (Years)",
       y = "Monthly Salary",
       fill = "Count") +
  theme_minimal()
```

Output:



13.Line chart

R Code:

```
library(ggplot2)
line_data <- aggregate(Monthly_Salary ~ Experience_Years,
                        employee,
                        mean)
ggplot(line_data,
       aes(x = Experience_Years,
           y = Monthly_Salary)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "red", size = 3) +
  labs(title = "Average Monthly Salary by Experience",
       x = "Experience (Years)",
       y = "Average Monthly Salary") +
  theme_minimal()
```

Output:



14.Pie chart

R Code:

```
library(ggplot2)
pie_data <- as.data.frame(table(employee$Department))
colnames(pie_data) <- c("Department", "Count")
ggplot(pie_data,
       aes(x = "",
```

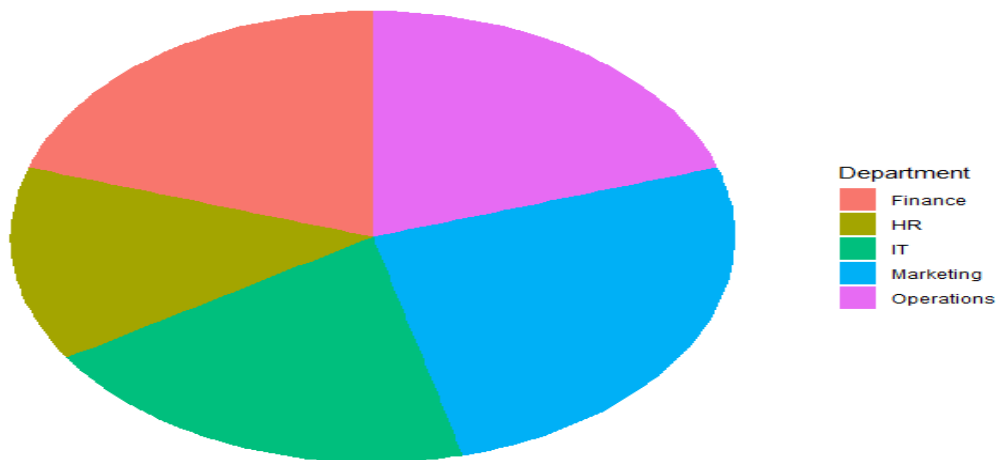
```

y = Count,
fill = Department)) +
geom_bar(stat = "identity", width = 1) +
coord_polar("y", start = 0) +
labs(title = "Employee Distribution by Department") +
theme_void()

```

Output:

Employee Distribution by Department



15.Area chart

R Code:

```

library(ggplot2)
area_data <- aggregate(Monthly_Salary ~ Experience_Years,
                        employee,
                        mean)
ggplot(area_data,
       aes(x = Experience_Years,
           y = Monthly_Salary)) +
geom_area(fill = "skyblue", alpha = 0.7) +
labs(title = "Area Chart of Average Salary by Experience",
     x = "Experience (Years)",
     y = "Average Monthly Salary") +
theme_minimal()

```

Output:

